

## Unit 2, Lesson 1: Square Values

### Benchmark

**MA.8.AR.2.3** Given an equation in the form of  $x^2 = p$  and  $x^3 = q$  where  $p$  is a whole number and  $q$  is an integer, determine the real solutions.

### Learning Target

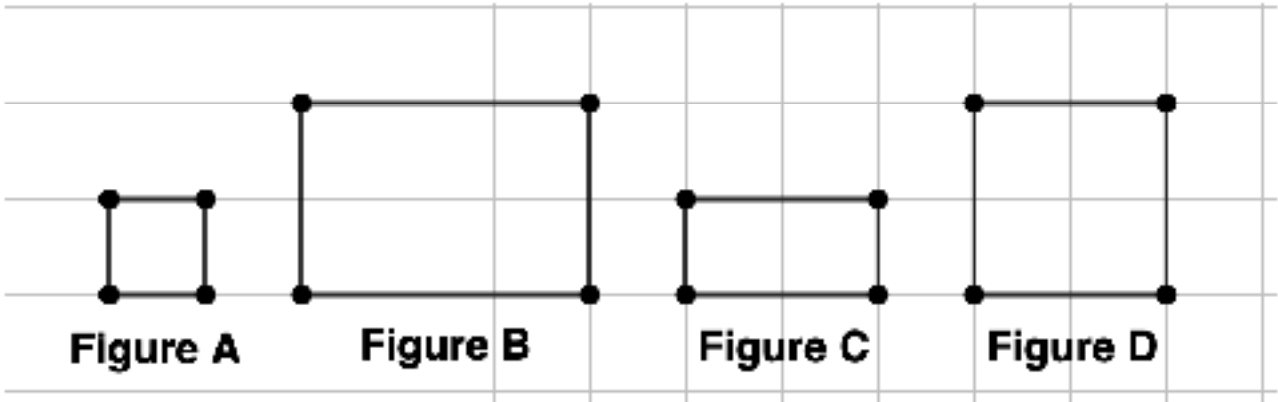
☐ I can use multiple representations to explore square values.



2.1.2 Exploration: Area of Squares

Complete the following with your partner.

1. Four figures are shown on the grid, where each gridline is 1 unit.

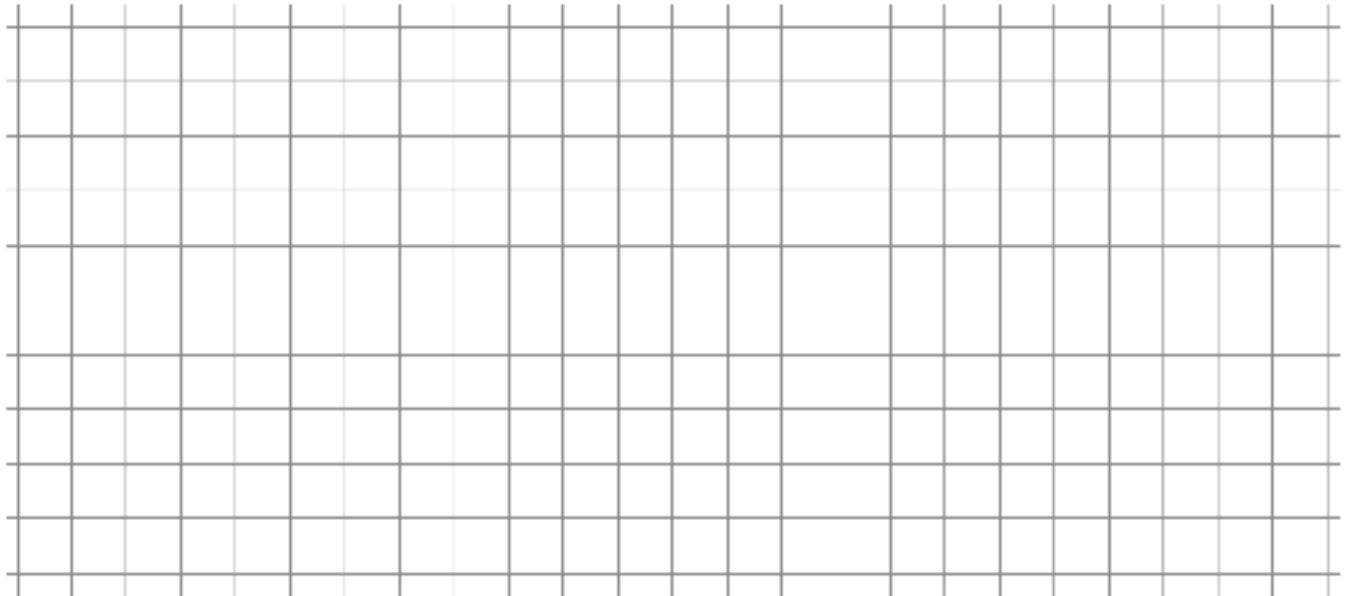


- a. Find the dimensions and area of each figure.

| Figure | Dimensions | Area (units <sup>2</sup> ) |
|--------|------------|----------------------------|
| A      |            |                            |
| B      |            |                            |
| C      |            |                            |
| D      |            |                            |

- b. All of the figures above are rectangles. Explain, using the dimensions, which of the shapes are also squares.

2. With your partner, create squares whose areas range from 1 to 25 square units. Create as many as possible on the grid paper below. Each gridline has a length of 1 unit.
- a. For each square, label the length and the width, along with the area inside.

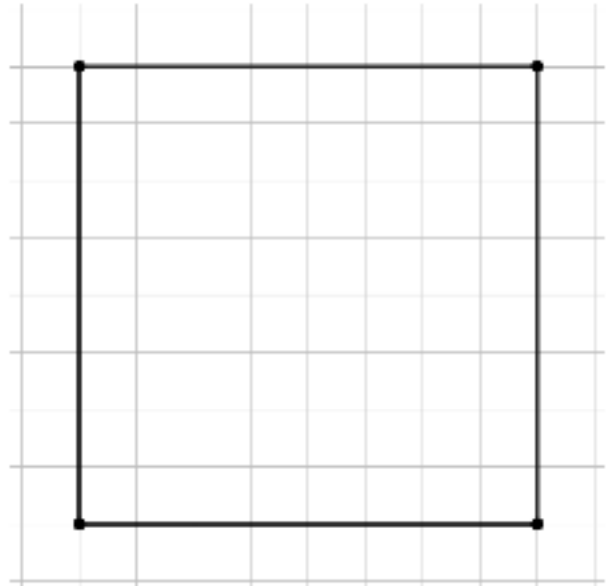


- b. Form a group with another pair of partners. Compare the squares each group created and, if necessary, add any squares to your grid.
3. Alicia says, "1 squared is 1 and 2 squared is 4." Explain what you think it means to square a number.

### 2.1.3 Bring It Together: Area of Squares

1. The area of the square shown is 64 square units.

- a. Label the length and the width of the square on the grid.
- b. The length and width can be multiplied to find the area of 64 square units. Since the figure is a square, the area can also be found using  $A = s^2$ , where  $s$  is the length of a side.

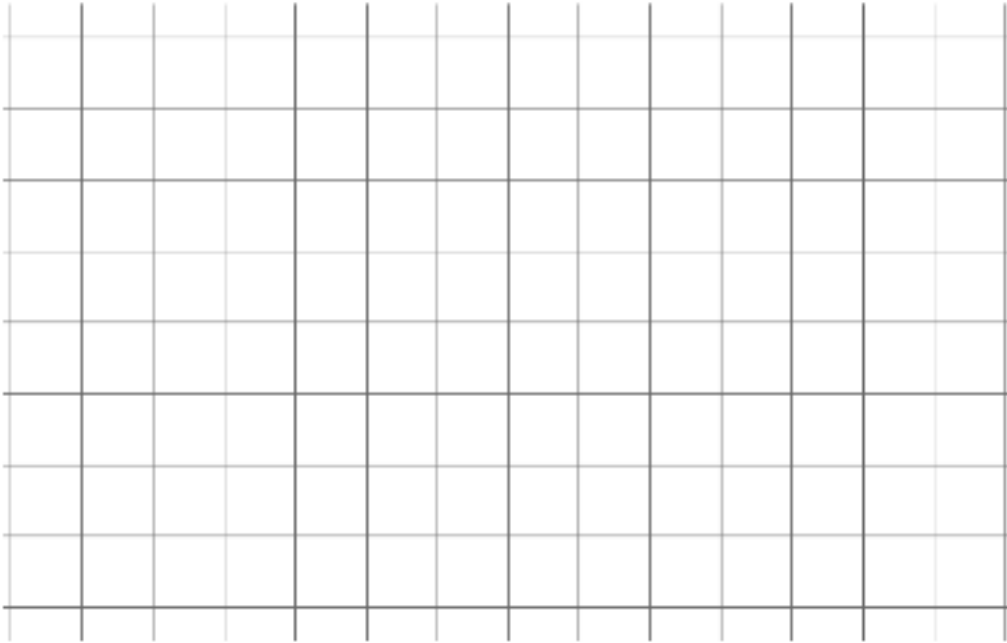


Complete the equations for the square shown.

$$\text{Area} = (\underline{\hspace{1cm}})^2 \quad \text{or} \quad \text{Area} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

2. Consider a square with an area of 100 units<sup>2</sup>.
- a. What is the length of each side?
- b. Explain how you determined the length.

- 3. Consider a rectangle with an area of 8 square units.**
- a. Use the grid to draw all of the possible rectangles with an area of 8 square units.**



- b. Explain whether it is possible to have a square with an area of 8 square units.**

4. The numbers 1, 4, 16, 25, 64, and 100 are all examples of perfect squares. The numbers 3, 8, and 200 are nonexamples of perfect squares.
- a. Discuss with your partner what it means for a number to be a perfect square. Then, write your definition of a perfect square.
- b. Ask two other pairs of partners the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

| My Summary of the Definition of a Perfect Square | Initials |
|--------------------------------------------------|----------|
|                                                  |          |
|                                                  |          |

- c. Review the definitions with your partner and make any necessary adjustments to your own definition.

### 2.1.4 Exploration: Square Values on the Coordinate Plane

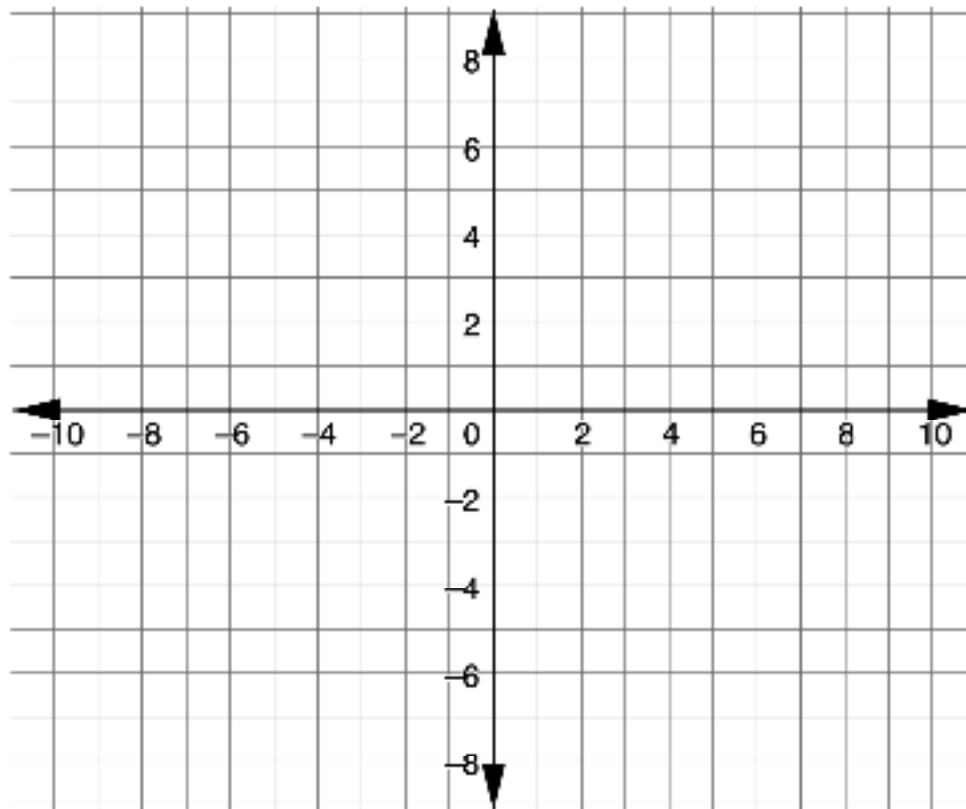
1. Use the equation  $y = x^2$  to complete the following with your partner.
  - a. Complete the table of values for the equation  $y = x^2$  by substituting the given values of  $x$  and solving for the corresponding  $y$ -value.

| $x$ | Substitute into<br>$y = x^2$                   | Coordinates<br>( $x, y$ ) |
|-----|------------------------------------------------|---------------------------|
| -3  | $y = (-3)^2$<br>$y = \underline{\hspace{2cm}}$ | <b>(-3, _____)</b>        |
| -2  |                                                |                           |
| -1  |                                                |                           |
| 0   |                                                |                           |
| 1   |                                                |                           |
| 2   |                                                |                           |
| 3   |                                                |                           |

- b. Discuss with your partner why the  $y$ -values are all perfect squares. Summarize your discussion.



- c. Plot the coordinates on the grid shown.



- d. The graph of  $y = x^2$  does not go below the  $x$ -axis. Complete the statements.

A positive number multiplied by itself results in a

- ☐ positive
- ☐ negative

product. A negative number multiplied by itself results in a

- ☐ positive
- ☐ negative

product. Therefore, for any value of  $x$  multiplied by itself

other than 0, the value of  $y$  will

- ☐ sometimes
- ☐ always

be

- ☐ positive,
- ☐ negative,

so all

of the points on the graph of  $y = x^2$  other than (0,0) are above the  $x$ -axis.

2. Four coordinate pairs are shown. The points represented are also on the graph of  $y = x^2$ .

$$(1.5, 2.25)$$

$$(-1.5, 2.25)$$

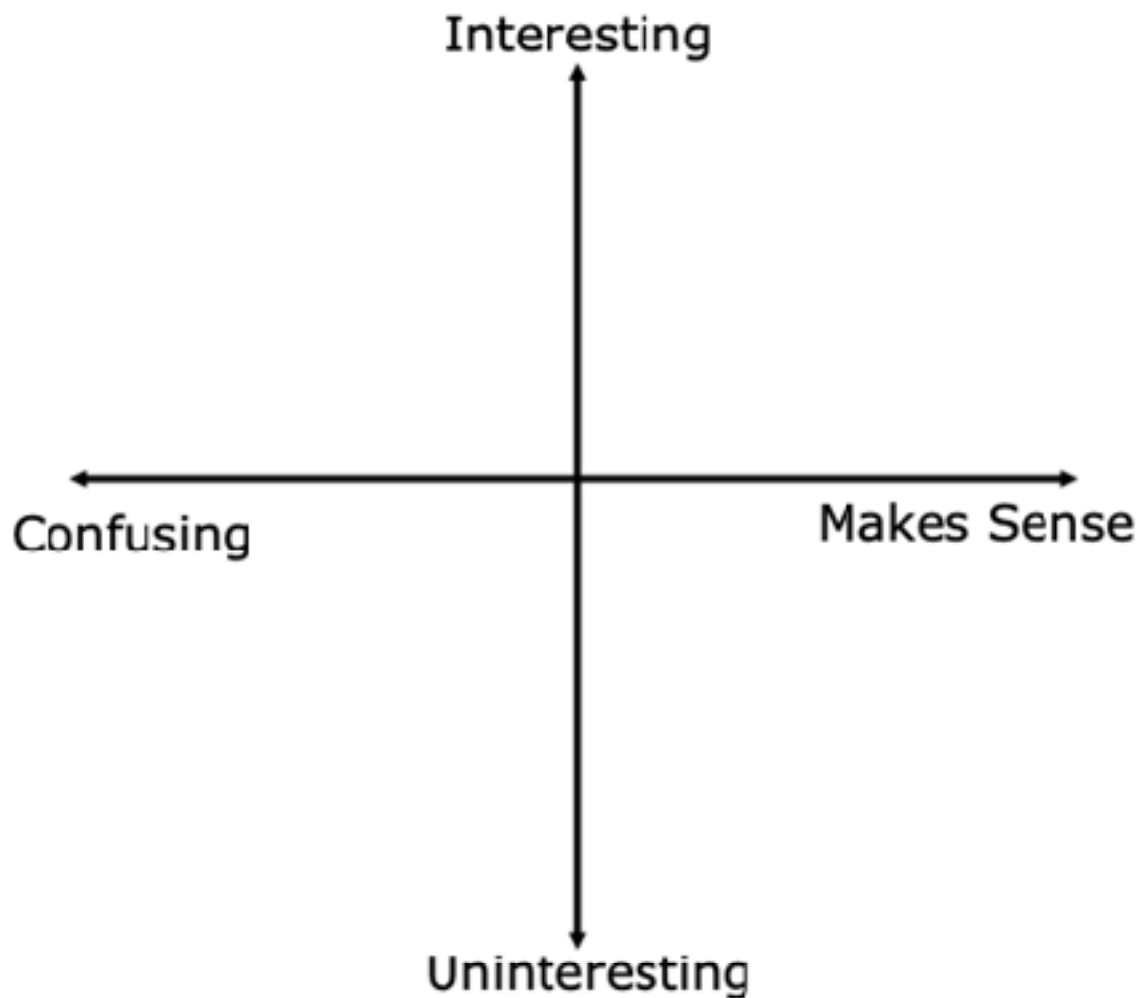
$$\left(\frac{5}{2}, \frac{25}{4}\right)$$

$$\left(-\frac{5}{2}, \frac{25}{4}\right)$$

- a. Locate and plot the given points on the graph.
- b. Discuss with your partner what you notice about the given coordinate pairs. Then, summarize your discussion.
- c. Choose one of the points and show how it satisfies the equation  $y = x^2$ .

### 2.1.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding of today's learning targets.



2. Write one question you have about square values after today's lesson.

